EXCEL DATA ANALYSIS SERIES

Excel PivotTable & PivotChart

Practical Learning Guide: from your first PivotTable to an interactive dashboard

1. Introduction to PivotTables and PivotCharts

What is a PivotTable?

A **PivotTable** is an Excel tool that turns a long list of records into a small, organized summary. You choose which categories to group by and which numbers to calculate; Excel does the adding, counting and averaging. No formulas are needed and your original data is never changed.

Why do Data Analysts use PivotTables?

- **Speed:** summarize thousands of rows in seconds by dragging fields.
- **Flexibility:** switch the view (by region, then by product) with a few clicks.
- **Reliability:** totals update when you refresh, with no formulas to maintain.
- **Insight:** patterns such as the best product or the weakest region become visible.

What is a PivotChart?

A **PivotChart** is a chart linked to a PivotTable. When you filter, regroup or refresh the PivotTable, the chart changes with it. Use the PivotTable to calculate the numbers and the PivotChart to show the pattern.

Normal table vs PivotTable vs PivotChart

	Normal table (raw data)	PivotTable	PivotChart
Purpose	Stores every record	Summarizes records	Visualizes the summary
What you see	One row per transaction	A small summary grid	Column, bar, line or pie chart
Calculates?	No, it only stores data	Yes: Sum, Count, Average, Min, Max	Uses the PivotTable's results
Our example	Each sale: date, region, revenue	Revenue by region	Column chart of revenue by region

From raw data to business insight

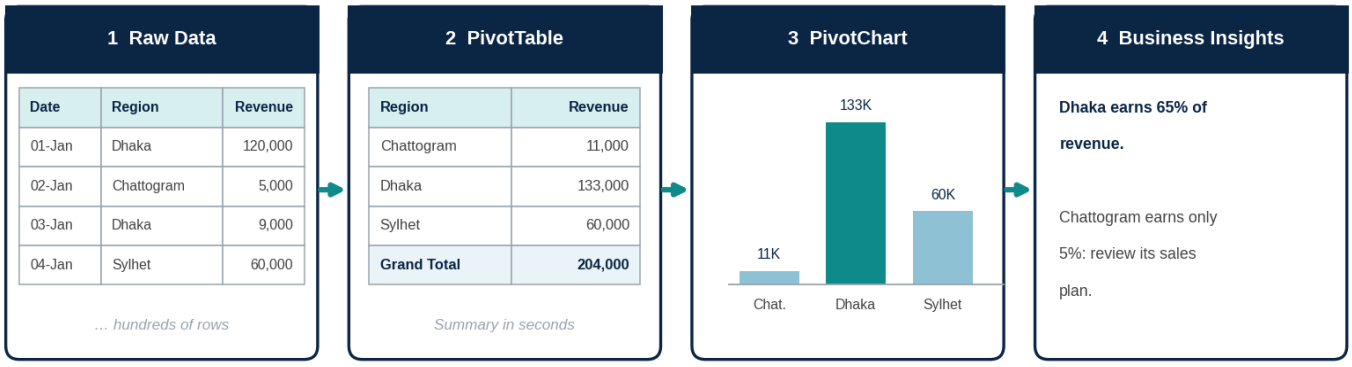


Figure 1. The analysis workflow. The same idea works for 6 rows or 600,000 rows.

Mini example: imagine hundreds of sales rows. A PivotTable with Region in Rows and Revenue in Values reduces them to three lines, and the chart shows at a glance that **Dhaka produces 133,000 of 204,000 revenue (65.2%)**.

Real-world business use cases

Use case	Business question	Typical fields
Sales analysis	Which days or months sell the most?	Date, Revenue
Product performance	Which product earns the most revenue?	Product, Units, Revenue
Monthly revenue	Is revenue growing month by month?	Date (grouped by month), Revenue
Regional performance	Which region needs attention?	Region, Revenue
Customer analysis	Who are our top buyers?	Customer, Orders, Revenue

HOW TO USE THIS GUIDE One sample dataset of six sales records (shown on Page 2) is used from start to finish. Every number in the tables, charts and answer key is calculated from it, so you can follow along in Excel and check your results.

2. Understanding the Four PivotTable Areas

When you create a PivotTable, Excel shows the **PivotTable Fields** pane. Every column in your data becomes a field. You build a report by dragging fields into four areas.

Area	What it does	Example (our data)	Question it answers
Rows	Categories listed down the side	Region	What is the revenue for each region?
Columns	Categories listed across the top	Product	How does each region split by product?
Values	Numbers to calculate or summarize	Sum of Revenue	How much did we earn in total?
Filters	Controls that filter the whole report	Salesperson	What do Amina's sales look like on their own?

The PivotTable Fields pane in action

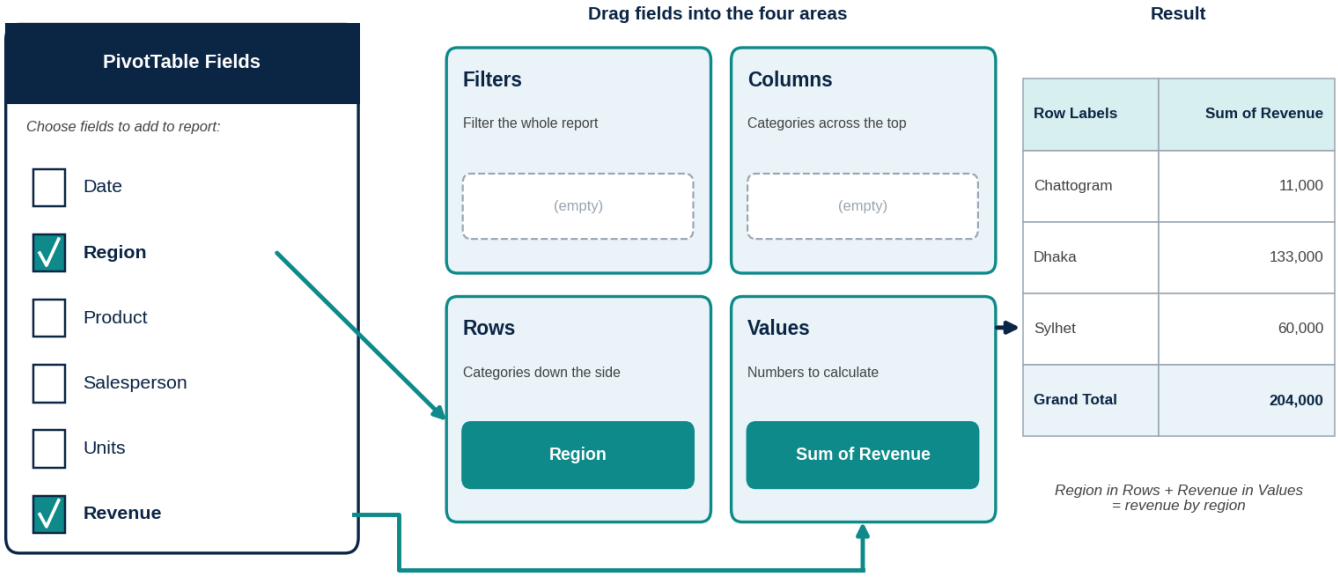


Figure 2. Dragging Region to Rows and Revenue to Values produces a regional revenue summary.

Sample dataset used throughout this guide

Date	Region	Product	Salesperson	Units	Revenue
01-Jan	Dhaka	Laptop	Amina	2	120,000
02-Jan	Chattogram	Mouse	Rahim	5	5,000
03-Jan	Dhaka	Keyboard	Sara	3	9,000
04-Jan	Sylhet	Laptop	Amina	1	60,000
05-Jan	Dhaka	Mouse	Rahim	4	4,000
06-Jan	Chattogram	Keyboard	Sara	2	6,000
Total				**17**	**204,000**

Enter dates as real Excel dates (for example 01-Jan-2026) so they can be grouped later.

Using Columns: a two-way summary

Add **Product** to Columns and you get a cross-tab. Each cell shows revenue for one region and one product (blank = no sales).

Sum of Revenue	Keyboard	Laptop	Mouse	Grand Total
Chattogram	6,000		5,000	11,000
Dhaka	9,000	120,000	4,000	133,000
Sylhet		60,000		60,000
Grand Total	**15,000**	**180,000**	**9,000**	**204,000**

> **BUSINESS INSIGHT** Dhaka brings in 133,000 of 204,000 (65.2%), mostly from laptops. Chattogram brings in only 11,000. The next step is to ask why.

3. Creating Your First PivotTable

Follow these steps with the sample dataset. Each step also says **why** it matters.

1. **Prepare a clean dataset.** One header row, one record per row, one type of data per column. Headers: Date, Region, Product, Salesperson, Units, Revenue.
2. **Remove blank rows and check consistency.** Delete empty rows or columns and fix spellings ("Dhaka", "dhaka" and "Dhaka " can appear as different regions). Make sure numbers are numbers and dates are real dates.
3. **Click inside the dataset.** Select any single cell; Excel detects the whole range automatically.
4. **Select Insert > PivotTable** (From Table/Range).
5. **Choose the table or range and destination.** Confirm the range (for example Sheet1!$A$1:$F$7), choose **New Worksheet**, then click OK.
6. **Drag fields into the areas.** Region to Rows, Product to Columns, Revenue to Values, Salesperson to Filters.
7. **Review the output.** Check that the Grand Total (204,000) equals the total of the Revenue column in your data.
8. **Rename the worksheet and format the result.** Rename the sheet (for example "Region Report"), then right-click a value > Number Format > Number with comma separator, and pick a style on the Design tab.

Key steps at a glance

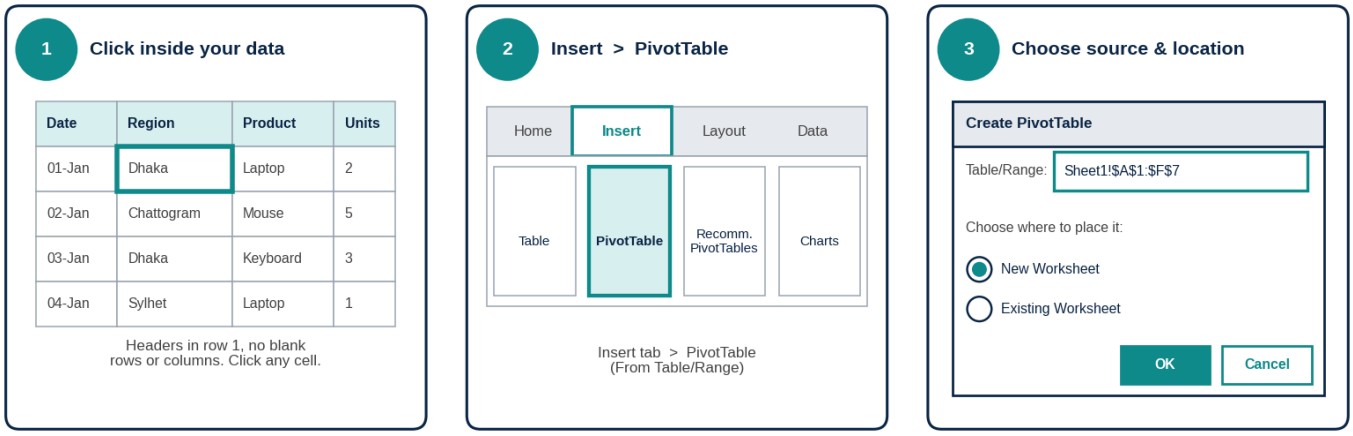


Figure 3. Excel-style illustration of steps 3 to 5.

Common mistakes and how to avoid them

Mistake	What happens	How to avoid it
Selecting an incomplete range	Some rows or columns are missing from the report	Click one cell inside the data (or use an Excel Table) and check the range in the dialog box
Blank column headers	Excel shows an error: the PivotTable field name is not valid	Give every column a unique, non-blank header before inserting
Revenue summarized as Count	Shows 6 (number of rows) instead of 204,000	Right-click the value > Summarize Values By > Sum. Count appears when a column holds blanks or text

WHY IT MATTERS A PivotTable only summarizes what it sees. Dirty data (extra spaces, blank headers, text numbers) produces wrong totals, so clean first and pivot second.

4. Summarizing and Analyzing Data

Right-click any number in the Values area and choose **Value Field Settings**. The **Summarize Values By** tab changes the calculation; the **Show Values As** tab changes how the result is displayed.

Summarize Values By: five core calculations

Function	Business question	Fields used	Result (sample data)
Sum	How much revenue did each region generate?	Rows: Region, Values: Revenue	Dhaka 133,000
Count	How many transactions did each product have?	Rows: Product, Values: Revenue	Laptop 2, Mouse 2, Keyboard 2
Average	What is the average sale per salesperson?	Rows: Salesperson, Values: Revenue	Amina 90,000; Sara 7,500; Rahim 4,500
Min	What is the smallest single sale?	Values: Revenue	4,000 (Dhaka, Mouse)
Max	What is the largest single sale?	Values: Revenue	120,000 (Dhaka, Laptop)

The overall average per transaction is 204,000 / 6 = **34,000**. Use Sum for totals, Count for activity, Average for typical size, and Min/Max for the range.

Summarizing vs displaying: before and after

Region	Sum of Revenue (original values)	% of Grand Total (display only)	Difference From Chattogram (display only)
Chattogram	11,000	5.4%	(base item)
Dhaka	133,000	65.2%	+122,000
Sylhet	60,000	29.4%	+49,000
Grand Total	**204,000**	**100.0%**	

KEY DIFFERENCE **Summarize Values By** changes what Excel calculates from the source rows (Sum, Count, Average, Min, Max). **Show Values As** keeps that calculation but re-displays the result relative to something else, such as a share of the total or a gap from a base item. The underlying numbers do not change.

Show Values As: when to use it

- **% of Grand Total:** "What share of revenue comes from each region?" Insight: Dhaka contributes 65.2%, so the business depends heavily on one region.
- **Difference From:** "How far ahead is each region compared with a benchmark?" Choose Base field = Region and Base item = Chattogram: Dhaka is 122,000 higher, Sylhet 49,000 higher.

Sorting and formatting

Task	How	Before > After
Sort largest to smallest	Right-click a value > Sort > Sort Largest to Smallest	Chattogram, Dhaka, Sylhet > Dhaka 133,000, Sylhet 60,000, Chattogram 11,000
Number / currency format	Value Field Settings > Number Format > Number or Currency	133000 > 133,000 (Number) or a currency symbol with 133,000.00
Percentage format	Show Values As > % of Grand Total, then Number Format > Percentage	0.652 > 65.2%

TIP Format values through Value Field Settings > Number Format, not by formatting cells, so the format survives when the PivotTable refreshes.

5. Grouping, Filtering, and Interactive Reports

Business question: "How is monthly sales performance changing, and what does it look like for one region?" Grouping organizes the data; filtering and slicers let readers explore it.

Grouping

- **Dates:** put Date in Rows, right-click a date > **Group**, then select Months, Quarters and/or Years. Dates must be valid Excel dates, not text. After adding the 07-Feb record from Exercise 9, the report shows Jan 204,000, Feb 12,000 and Qtr1 216,000.
- **Numbers:** put Revenue in Rows, right-click > Group, and enter Starting at 0, Ending at 150,000, By 50,000. Count the transactions in each band: 0-49,999 has 4, 50,000-99,999 has 1, 100,000-149,999 has 1. Insight: most sales are small; only two are large.

Sorting and filtering tools

Tool	What it does	Example (sample data)
Sort	Orders labels or values	Products by revenue: Laptop, Keyboard, Mouse
Report Filter	Filters the whole report from the Filters area	Salesperson = Amina: revenue is 180,000
Label Filter	Filters by the text of a label	Product begins with K: Keyboard only
Value Filter	Filters by calculated values	Regions with revenue greater than 50,000: Dhaka and Sylhet
Top 10	Keeps the top or bottom N items	Top 2 products by revenue: Laptop 180,000 and Keyboard 15,000
Slicer	Click-button filter for a field	PivotTable Analyze > Insert Slicer > Region
Timeline	Slider filter for a date field	Insert > Timeline > Date; choose Months or Quarters

Slicers in action

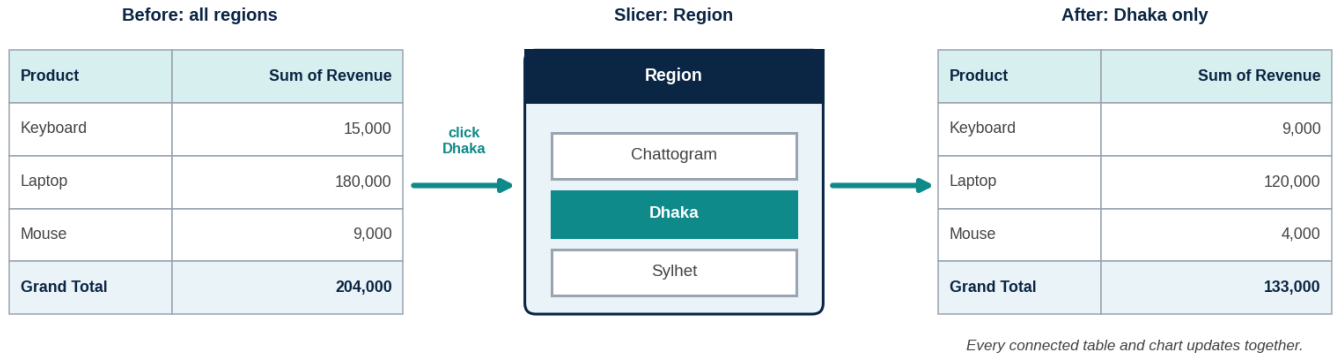


Figure 4. Selecting the Dhaka slicer button changes the PivotTable from 204,000 to 133,000.

Monthly performance workflow

1. Put Date in Rows and Revenue in Values, then group Date by Months (and Quarters).
2. Insert a **Region slicer** (PivotTable Analyze > Insert Slicer) and a **Timeline** (PivotTable Analyze > Insert Timeline).
3. Click Dhaka and check the monthly revenue; click Sylhet to compare. Click the Clear Filter icon in the slicer to reset.

> **COMPATIBILITY** Date grouping needs valid Excel dates (a column of text dates will not group). Slicers and Timelines are available in supported Excel versions and configurations; if your Insert Slicer button is greyed out, check your Excel version and that you clicked inside a PivotTable.

6. Creating and Formatting PivotCharts

A **PivotChart** is linked to a PivotTable: changing one changes the other. **To create one:** click inside the PivotTable, then choose **Insert > PivotChart** (or PivotTable Analyze > PivotChart), pick a chart type and click OK. The chart's axis shows the Rows fields, the legend shows the Columns fields, and the plotted numbers come from Values.

Formatting a PivotChart

- **Change type:** right-click the chart > Change Chart Type (or PivotChart Analyze > Change Chart Type).
- **Elements:** Chart Design > Add Chart Element > Chart Title, Axis Titles, Data Labels, Legend. Label the axis with units (Revenue).
- **Number format:** right-click the axis or labels > Format Axis > Number (for example thousands with comma).
- **Filtering:** use the field buttons on the chart, or insert a Slicer and connect it with Report Connections so one click updates chart and table.

Five chart types for five business questions

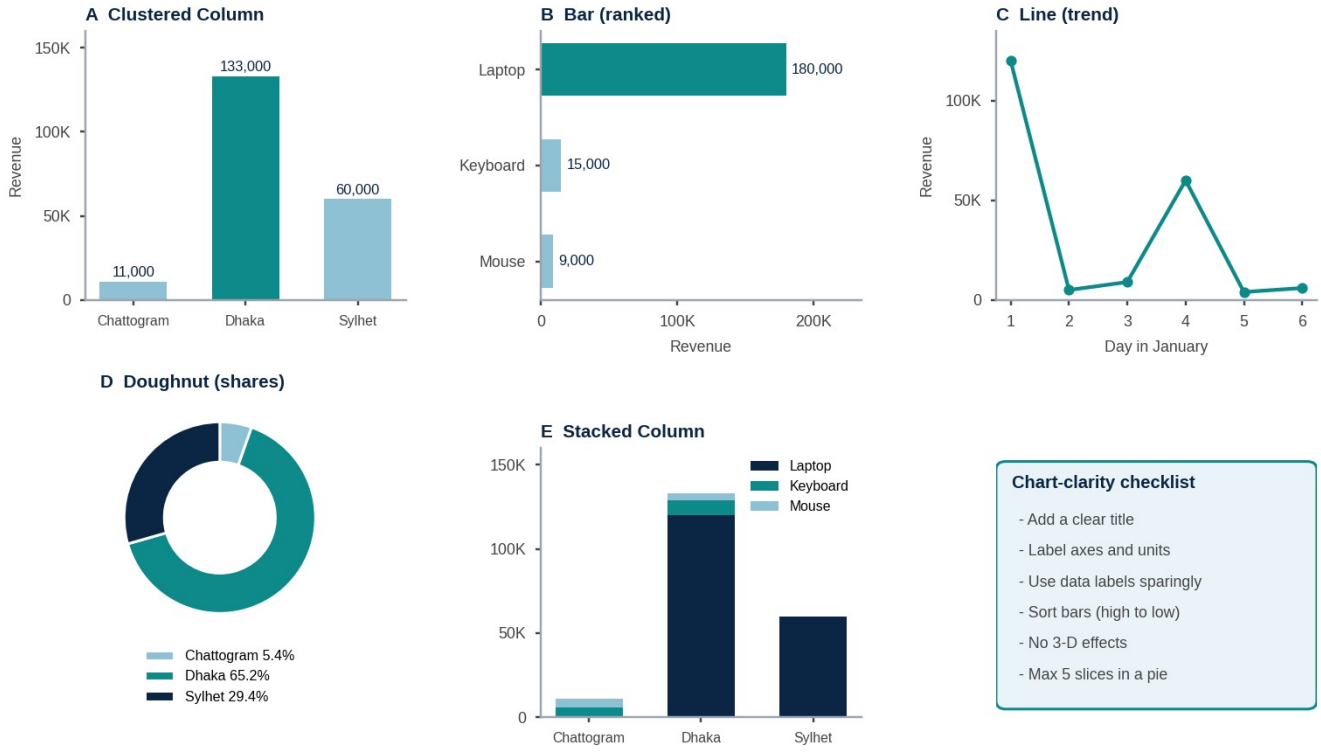


Figure 5. Five chart types built from the sample dataset (C shows daily revenue, 1 to 6 January).

Chart type	Best for	Business question	Insight (sample data)
Clustered Column (A)	Comparing categories	Which region earns the most?	Dhaka leads with 133,000
Bar (B)	Ranking items	Which product performs best?	Laptop 180,000 far ahead of Keyboard 15,000 and Mouse 9,000
Line (C)	Trends over time	How did revenue move from 1 to 6 Jan?	Peaks on 1 Jan (120,000) and 4 Jan (60,000)
Pie / Doughnut (D)	Shares of a whole (few categories)	What share does each region contribute?	Dhaka 65.2%, Sylhet 29.4%, Chattogram 5.4%
Stacked Column (E)	Composition within categories	What makes up each region's revenue?	Dhaka: Laptop 120,000 of 133,000

DESIGN RULE Avoid 3-D effects and clutter. Use one highlight colour for the main message, sort bars, and use a pie or doughnut only for a small number of categories (five or fewer).

7. Build a Mini Sales Dashboard

Goal: one worksheet with two KPI cards, three charts and a Region slicer, all built from the sample dataset.

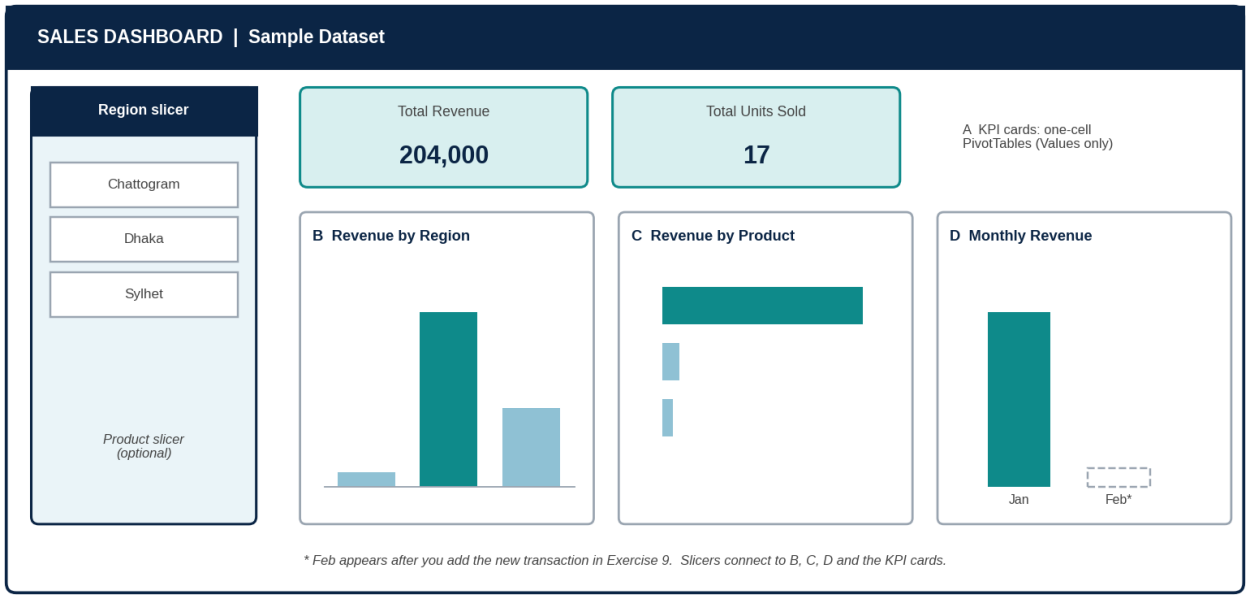


Figure 6. Dashboard wireframe. Each block maps to one PivotTable or chart.

Step-by-step build

1. **Create the PivotTables** (one per component, on a sheet called Calc): see the table below.
2. **Insert the PivotCharts:** click inside the Region table > Insert > PivotChart > Clustered Column; Product table > Bar; Month table > Column (or Line).
3. **Arrange on one worksheet:** add a sheet named Dashboard; cut and paste the charts and KPI PivotTables into the layout above. Leave space for slicers.
4. **Add slicers and connect them:** click a PivotTable > PivotTable Analyze > Insert Slicer > Region. Right-click the slicer > **Report Connections** and tick every PivotTable on the dashboard.
5. **Apply consistent formatting:** same colours and fonts, chart titles, number formats; hide gridlines (View > Gridlines) and hide chart field buttons (PivotChart Analyze > Field Buttons).
6. **Test the filters** (see the check table below).

PivotTable	Rows	Values	Feeds
PT1 Region	Region	Sum of Revenue	Revenue by Region chart
PT2 Product	Product	Sum of Revenue	Revenue by Product chart
PT3 Month	Date (grouped by Months)	Sum of Revenue	Monthly Revenue chart
PT4 / PT5 KPIs	(none)	Sum of Revenue / Sum of Units	Total Revenue and Total Units cards

Does it work? Expected results when you click a slicer button

Slicer selection	Total Revenue	Total Units	Product chart should show
None (all regions)	204,000	17	Laptop 180,000; Keyboard 15,000; Mouse 9,000
Dhaka	133,000	9	Laptop 120,000; Keyboard 9,000; Mouse 4,000
Chattogram	11,000	7	Keyboard 6,000; Mouse 5,000
Sylhet	60,000	1	Laptop 60,000

> **NOTE** A Region slicer also narrows the Revenue by Region chart to the selected region; that is normal. With only January data, the monthly chart shows one column until you add the February record in Exercise 9.

8. Intermediate to Advanced Techniques

These techniques extend what you learned. Feature availability depends on your Excel version and platform, so check your own installation before relying on a feature.

Technique	Purpose	Real-world use case	Notes / availability
Multiple fields in Rows and Columns	See detail across two or more categories	Region then Product: which products sell in each region?	Field order changes the layout
Nested categories and subtotals	Show details under a parent with totals	Dhaka total with its products beneath	Design > Subtotals / Grand Totals
Grouping dates into periods	Months, quarters, years	Monthly and quarterly revenue reports	Needs valid dates
Calculated Field	New field from existing fields	Average price = Revenue / Units = 204,000 / 17 = 12,000	PivotTable Analyze > Fields, Items & Sets. Not available for Data Model PivotTables (use a measure)
Distinct Count	Count unique items	How many different salespeople sold in each region? Dhaka 3, Chattogram 2, Sylhet 1	Tick Add this data to the Data Model when creating; support varies by version
Excel Table as source	Range grows automatically	A daily sales log that keeps growing	Select data, press Ctrl+T, then insert PivotTable
Refresh	Update after data changes	A new transaction was added	Right-click > Refresh, or PivotTable Analyze > Refresh
Change Data Source	Point the report to a new range	Source grew beyond the old range	PivotTable Analyze > Change Data Source
Power Query	Clean or combine data before pivoting	Merge monthly files, fix text, remove duplicates	Data > Get Data; built in to recent versions, add-in for some older ones

Example: nested categories with subtotals

Put **Region** and then **Product** in Rows. Excel lists products under each region and adds a subtotal.

Region / Product	Sum of Revenue	Share of Grand Total
Chattogram	11,000	5.4%
Keyboard	6,000	2.9%
Mouse	5,000	2.5%
Dhaka	133,000	65.2%
Keyboard	9,000	4.4%
Laptop	120,000	58.8%
Mouse	4,000	2.0%
Sylhet	60,000	29.4%
Laptop	60,000	29.4%
Grand Total	**204,000**	**100.0%**

Region rows are the subtotals; indented rows are the products inside each region.

Data Model and Calculated Fields: why availability matters

- A **Calculated Field** works on totals: Revenue / Units for the whole table is 12,000 per unit, not an average of averages.
- **Distinct Count** is only offered when the PivotTable is built on the Data Model. Using the Data Model changes which features are available, so use it when you need it.
- **Power Query rule of thumb:** use it first when data comes from several files, has messy text, or needs repeated cleanup. Use a plain Excel Table when the data is already clean.

> **BUSINESS QUESTION** "How many salespeople does each region rely on?" Distinct Count shows that Sylhet depends on one person (Amina), a risk the manager can act on.

9. Common Errors, Troubleshooting, and Best Practices

Problem	Likely cause	Solution
Shows Count instead of Sum	The Revenue column has blanks or text values	Fix the data, refresh, then Value Field Settings > Sum
New records missing after refresh	Source range is fixed and does not include new rows	Convert data to an Excel Table, or PivotTable Analyze > Change Data Source; then Refresh
Dates cannot be grouped	Dates are text, or the date column has blanks or invalid entries	Convert to real dates (Text to Columns or DATEVALUE), fill blanks, refresh, group again
Blank or incorrect field names	Missing or duplicate headers in the source	Give each column a unique header; refresh
Duplicate or inconsistent categories	Spelling, case or extra spaces (Dhaka / Dhaka)	Clean with Find and Replace or TRIM; then refresh
Numbers stored as text	Numbers imported with apostrophes or text format	Convert to Number (Text to Columns or Paste Special > Multiply by 1)
Slicers do not control every chart	Slicer is not connected to every PivotTable	Right-click slicer > Report Connections > tick all PivotTables
PivotChart formatting changes unexpectedly	Changing fields or filters rebuilds series and resets formatting	Finish the layout first, then format; keep the field layout stable; save a chart template
Results are wrong	Source data contains duplicates or errors	Remove duplicates (Data > Remove Duplicates), fix errors, then check totals against the raw data

Best practices for professional data analysis

1. **Keep raw data separate from reports.** Store data on one sheet and each report on its own sheet.
2. **Use clear field names.** One short, unique header per column (Revenue, not Column6).
3. **Avoid merged cells in source data.** Merged cells break PivotTables and filters.
4. **Format source data as an Excel Table** (Ctrl+T) so new rows are included automatically.
5. **Verify totals against the original data.** For the sample data, the Grand Total must equal 204,000.
6. **Refresh reports after source changes.** A PivotTable does not update on its own.
7. **Never assume a PivotTable result is correct without validating it.** Compare with a quick SUM, a count of rows, or a known figure.

> **VALIDATION HABIT** After every PivotTable, ask three questions: Does the Grand Total match the source? Does the row count look right? Do any categories look duplicated or misspelled?

10. Practical Exercises and Quick Revision

Part A: Exercises (use the sample dataset on Page 2)

#	Task	How to do it
1	Total revenue by region	Insert > PivotTable (New Worksheet). Region to Rows, Revenue to Values. Check it says Sum of Revenue.
2	Best-selling product by revenue	Product to Rows, Revenue to Values. Right-click a value > Sort > Sort Largest to Smallest. The top row is the answer.
3	Average revenue per transaction	Remove Region. Value Field Settings > Average. Cross-check: total revenue / number of rows.
4	Group sales by month	Date to Rows. Right-click a date > Group > Months. (Needs real dates.)
5	Each region's % of total revenue	Region to Rows, Revenue to Values. Value Field Settings > Show Values As > % of Grand Total.
6	PivotChart comparing products	Click inside the Product table > Insert > PivotChart > Clustered Column or Bar. Add title, axis title, data labels.
7	Region slicer	PivotTable Analyze > Insert Slicer > Region. Click each region and note what changes.
8	One-page mini dashboard	Follow Page 7: two KPI cards, three charts, one slicer connected with Report Connections.
9	Refresh after a new transaction	Add: 07-Feb | Sylhet | Keyboard | Sara | 4 | 12,000 to the Excel Table. PivotTable Analyze > Refresh. Note the new totals.
10	Two manager insights	Write two insights, each with a recommended action (use % and totals from your results).

Part B: Quick revision sheet

Term	Meaning	Term	Meaning
PivotTable	Interactive summary built from source data	Slicers	Click-button filters for PivotTables and PivotCharts
PivotChart	Chart linked to a PivotTable	Timelines	Slider filter for a date field
Rows	Categories listed down the side	Refresh	Update a report with the latest source data
Columns	Categories listed across the top	Filters	Filter the whole report from the Filters area
Values	Numbers summarized (Sum, Count, Average, Min, Max)	Data Model	Linked-table model that enables Distinct Count and measures (where supported)

ANSWER KEY (check only after you finish the exercises)	
1	Dhaka 133,000; Sylhet 60,000; Chattogram 11,000; Grand Total 204,000.
2	Laptop with 180,000 (88.2% of revenue); Keyboard 15,000; Mouse 9,000.
3	34,000 (204,000 / 6 transactions).
4	Original data: Jan 204,000. After Task 9: Jan 204,000 and Feb 12,000.
5	Dhaka 65.2%, Sylhet 29.4%, Chattogram 5.4%. After Task 9 (total 216,000): Dhaka 61.6%, Sylhet 33.3%, Chattogram 5.1%.
6	Laptop 180,000; Keyboard 15,000; Mouse 9,000 (after Task 9, Keyboard becomes 27,000).
7	Dhaka selected: revenue 133,000, 9 units; Laptop 120,000, Keyboard 9,000, Mouse 4,000.
8	KPIs: Total Revenue 204,000 and Total Units 17 (216,000 and 21 after Task 9).
9	After refresh: revenue 216,000; units 21; Sylhet 72,000; Keyboard 27,000; Feb 12,000.
10	Sample: (a) Dhaka gives 65.2% of revenue, so protect its customers and stock. (b) Laptops are 3 of 17 units (17.6%) but 88.2% of revenue, so focus on laptop sales and pricing; Chattogram at 5.4% needs a review.